习题9.1 1,2,5(1)(3)(4),7,10,11,12
习题9.2 3,4,5,7,8. 习题9.3 偶数题目
习题9.4 题目1,2,5,9，每道大题任选两道小题， 本节其余题目基本上都非常经典，建议全做（至少做 n-3 道）；
习题9.5 2,4,6,8,9; 习题9.6 4,6,7. 习题9.7 1-5大题每道题任选一道小题,10 题

9.1

9.1.1

(1)

$$\int_a^b \left| x - \frac{a+b}{2} \right| \mathrm{d}x = \frac{(a-b)^2}{4}.$$

(2)

$$\int_a^b \sqrt{(x-a)(b-x)}\,\mathrm{d}x = \int_a^b \sqrt{-\left(x - \frac{a+b}{2}\right)^2 + \frac{(a-b)^2}{4}}\,\mathrm{d}x = \frac{\pi(a-b)^2}{8}.$$

9.1.2

Suppose $f \in \mathcal{R}([a,b])$ prove that

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{n\to\infty} \frac{b-a}{n} \sum_{k=1}^n f\left(a + \frac{k}{n}(b-a)\right).$$

Proof: Let $x_k^{(n)} = a + \frac{k}{n}(b-a)$ then for all n, $x_0^{(n)}, \cdots, x_n^{(n)}$ is a partition of $[a,b]$ and $\frac{b-a}{n}\sum_{k=1}^n f(x_k^{(n)})$ is a Riemann sum. Since $x_{k+1}^{(n)} - x_k^{(n)} = \frac{b-a}{n} \to 0$, the sum converges to the integral $\int_a^b f(x)\,\mathrm{d}x$.

9.1.5

Determine the integrability of f on $[0,1]$:
(1) $f(x) = -\mathbf{1}_{\mathbb{Q}} + \mathbf{1}_{\mathbb{Q}^C}$: non-integrable.
(3) $f(x) = \operatorname{sgn}\left(\sin\frac{\pi}{x}\right)$, $f(0) = 0$: integrable.
(4) $f(x) = \left\{\frac{1}{x}\right\}$, $f(0) = 0$: integrable.

9.1.7

Suppose $f \in \mathcal{R}([a,b])$, and f, g agree on all but finitely many points. Prove that $g \in \mathcal{R}([a,b])$ and $\int_a^b f\,\mathrm{d}x = \int_a^b g\,\mathrm{d}x$.
Proof: g is continuous a.e. since it is continuous at all continuities of f except for finitely many points. $f - g = 0$ except for $x_1, \cdots, x_n$ so $\int_a^b f - g\,\mathrm{d}x = \sum_{k=0}^n \int_{x_k}^{x_{k+1}} f - g\,\mathrm{d}x = 0$.

9.1.10-11

Suppose $f \in \mathcal{R}([a,b])$ prove that $e^f \in \mathcal{R}([a,b])$. If $f \geqslant c > 0$, $\log f \in \mathcal{R}([a,b])$.
Proof: f is bounded implies e^f is bounded on $[a,b]$. If f is continuous at x then e^f is also continuous at x, so e^f is continuous a.e. Hence $e^f \in \mathcal{R}([a,b])$.
$\log f$ is analogous.

9.1.12

If f is bounded on $[a,b]$, and its set of discontinuities D_f has only finitely many limit points, then $f \in \mathcal{R}([a,b])$.
Proof: Suppose the limit points of D_f are $x_1, \cdots, x_n$, then for any $y \notin \{x_1, \cdots, x_n\}$ there is an $\varepsilon(y) > 0$ such that $B(y, \varepsilon(y)) \cap D_f \subset \{y\}$. For any $\delta > 0$, $\bigcup_{y \neq x_i} B(y, \varepsilon(y)) \cup \bigcup_{i=1}^n B(x_i, \delta)$ is an open covering of $[a,b]$. Since $[a,b]$ is compact, there is a finite sub-covering $\bigcup_{j=1}^m B(y_j, \varepsilon_j) \cup \bigcup_{i=1}^n B(x_i, \delta)$ of $[a,b]$. Hence for any $\delta = 1/n > 0$, only finitely many points of D_f lie outside $\bigcup_{i=1}^n B(x_i, \delta)$, denote them by A_n. Note that $D_f \subset \bigcup_{n \geqslant 1} A_n \cup \{x_1, \cdots, x_n\}$, so D_f is countable, and $f \in \mathcal{R}([a,b])$.

9.2

9.2.3

If $f^2 \in \mathcal{R}([a,b])$, then $|f| \in \mathcal{R}([a,b])$.
Proof: $|f|$ is continuous at all continuities of f^2.

9.2.4

Suppose $f \in \mathcal{R}([a,b])$, if $f > 0$ then $\int_a^b f \,\mathrm{d}x > 0$.
Proof: For any partition of f, the lower Darboux sum is strictly positive, so the integral is positive.

9.2.5

Suppose $A \subset \mathbb{R}$. If A is a closed null set, prove that χ_A is integrable on any closed interval and the integral is zero.
Proof: $\chi_A = 0_{\mathbb{R}}, a.e.$ so the integral is zero.

9.2.7

Suppose $f, g \in \mathcal{R}([a,b])$. If f, g disagree on only a null set, prove that their integral are the same.
Proof: We show that if $f = 0, a.e.$ then $\int_a^b f \,\mathrm{d}x = 0$:
Suppose $|f| \leqslant M$ on $[a,b]$, then for all $\varepsilon > 0$, $\{x : f(x) \neq 0\}$ can be covered by intervals $\bigcup_{i=1}^n (l_i, r_i)$ with length less than $\delta = \varepsilon / M$. Then $\int_a^b f \,\mathrm{d}x = \sum_{i=1}^n \int_{l_i}^{r_i} f \,\mathrm{d}x \leqslant M\delta \leqslant \varepsilon$ and likewise $\int_a^b f \,\mathrm{d}x \geqslant -\varepsilon$. Therefore $\int_a^b f \,\mathrm{d}x = 0$.

9.2.8

Suppose $f \in \mathcal{R}([a,b])$, g is bounded and disagree with f only on a null set, show that $g \in \mathcal{R}([a,b])$.
Proof: The union of two null sets are still null sets, and if f is continuous at x and $f(x) = g(x)$, then g is continuous at x.

9.3

9.3.2

Give an example of F differentiable on $[a,b]$ but F' not integrable on $[a,b]$.
Solution: Consider $F:[0,1]\to\mathbb{R}, x\mapsto x^2\sin\left(1/x^2\right), F(0)=0$, then $F'(0)=0$ but $F'(x)=2x\sin\left(1/x^2\right)-\frac{2}{x}\cos\left(1/x^2\right)$ is unbounded near $x=0$, hence F' is not Riemann integrable.

9.3.4

Calculate
(1)

$$\int_0^1 x^2\,\mathrm{d}x=\frac{1}{3}.$$

(2)

$$\int_a^b \sin x\,\mathrm{d}x=\cos a-\cos b.$$

(3)

$$\int_0^\pi \cos^2 x\,\mathrm{d}x=\frac{\pi}{2}.$$

9.3.6

Suppose $f\in C([a,b])$ and $\int_a^b f^2\,\mathrm{d}x=0$, prove that $f\equiv 0$.
Proof: Otherwise if $f(x_0)\neq 0$, suppose $x_0\neq b$ and let $\varepsilon=|f(x_0)|/2$, then there exists $\delta>0$ such that $|f(y)|>\varepsilon\forall y\in[x_0,x_0+\delta]$. Hence $\int_a^b f^2\,\mathrm{d}x\geqslant\int_{x_0}^{x_0+\delta} f^2\,\mathrm{d}x>\varepsilon^2\delta>0$ a contradiction. Hence $f\equiv 0$.

9.3.8

Suppose $f\in C^1([a,b])$ and $f(a)=0$, prove that

$$\int_a^b f^2\,\mathrm{d}x\leqslant\frac{(b-a)^2}{2}\int_a^b|f'|^2\,\mathrm{d}x.$$

Proof: Note that

$$f(x)^2=\left(\int_a^x f'\,\mathrm{d}x\right)^2\leqslant\int_a^x|f'|^2\,\mathrm{d}x\int_a^x 1^2\,\mathrm{d}x=(x-a)\int_a^b|f'|^2\,\mathrm{d}x.$$

Hence

$$\int_a^b f^2\,\mathrm{d}x\leqslant\int_a^b|f'|^2\,\mathrm{d}x\int_a^b x-a\,\mathrm{d}x=\frac{(b-a)^2}{2}\int_a^b|f'|^2\,\mathrm{d}x.$$

9.3.10

Calculate the derivatives of F:
(1) $F(x) = \int_0^{\cos x^2} \sqrt{1+t^2}\,\mathrm{d}t$; (2) $F(x) = \int_{\sin x}^{x} \mathrm{d}t/\sqrt{1+\sin^2 t}$; (3) $F(x) = \int_a^b \sin(x+t)\,\mathrm{d}t$.
Solution: (1)

$$F'(x) = -2x\sin x^2\sqrt{1+\cos^2 x^2}.$$

(2)

$$F'(x) = \sqrt{1+\sin^2 x} - \cos x/\sqrt{1+\sin^2 \sin x}.$$

(3)

$$F'(x) = \frac{d}{dx}\int_{x+a}^{x+b} \sin t\,\mathrm{d}t = \sin(x+b) - \sin(x+a).$$

9.3.12

Suppose $f \in C(0,\infty)$. If

$$\int_0^x f(t)\,\mathrm{d}t = \frac{1}{2}xf(x), \forall x > 0,$$

prove that there is a constant c such that $f(x) = cx$.
Proof: Let $F(x) = \int_0^x f\,\mathrm{d}t$, then $F(x) = \frac{1}{2}xf(x)$. $f(x) = 2F(x)/x$ is differentiable, so $f(x) = F'(x) = \frac{1}{2}f(x) + \frac{1}{2}xf'(x)$ and $f(x) = xf'(x)$. Hence $\frac{d}{dx}\frac{f(x)}{x} = 0$ so $f(x) = cx$ for some constant c.

9.3.14

Suppose f is convex on $[a,b]$, prove that $f \in \mathcal{R}([a,b])$ and

$$(b-a)f\left(\frac{a+b}{2}\right) \leqslant \int_a^b f\,\mathrm{d}x \leqslant (b-a)\frac{f(a)+f(b)}{2}.$$

Proof: f is convex implies f is continuous (differentiable) except for finitely many points, and f is bounded by $f(a), f(b)$, so $f \in \mathcal{R}([a,b])$. Note that if $x+y = a+b$ then $f(x)+f(y) \geqslant 2f((a+b)/2)$ and $f(x)+f(y) \leqslant f(a)+f(b)$, hence

$$2(b-a)f\left(\frac{a+b}{2}\right) \leqslant 2\int_a^b f\,\mathrm{d}x \leqslant (b-a)(f(a)+f(b)).$$

9.4

9.4.1

(5)

$$\int_0^1 x\sqrt{1-x}\,\mathrm{d}x = \int_0^1 (1-x)\sqrt{x}\,\mathrm{d}x = \frac{2}{3} - \frac{2}{5} = \frac{4}{15}.$$

9.4.2

(10)

$$\int_0^1 \frac{\sqrt{1+x^2}-1}{x^2}\,\mathrm{d}x = \int_0^{\pi/2} \frac{1/\cos^2\theta - 1}{\sin^2\theta}\,\mathrm{d}\theta = \int_0^{\pi/2} \frac{1}{\cos\theta} - \frac{1}{1+\cos\theta}\,\mathrm{d}\theta = \log\left(\sqrt{2}+1\right) - \sqrt{2} + 1.$$

9.4.3

Suppose $0 < r < 1$, prove that

$$\int_0^{2\pi} \frac{1-r^2}{1-2r\cos x + r^2}\,\mathrm{d}x = 2\pi.$$

Proof: The Poisson kernel

$$K_r(x) = \frac{1-r^2}{1-2r\cos x + r^2} = 1 + 2\sum_{n\geqslant 1} r^n \cos nx.$$

Since $K_r(x)$ converges uniformly and $\int_0^{2\pi} \cos nx\,\mathrm{d}x = 0$,

$$\int_0^{2\pi} K_r(x)\,\mathrm{d}x = 2\pi.$$

9.4.4

Suppose $n \in \mathbb{N}$, prove that

$$\int_0^1 \frac{\mathrm{d}x}{(1+x^n)\sqrt[n]{1+x^n}} = 2^{-1/n}.$$

Proof: Let $u = 1 + x^{-n}$, then

$$\int_0^1 \frac{\mathrm{d}x}{(1+x^n)^{(n+1)/n}} = \int_0^1 x^{-(n+1)}(1+x^{-n})^{-(n+1)/n}\,\mathrm{d}x = \frac{1}{n}\int_2^\infty u^{-(n+1)}\,\mathrm{d}u = 2^{-1/n}.$$

9.4.5

(3)

$$\int_{-\pi}^{\pi} \sin mx \cos nx\,\mathrm{d}x = \frac{1}{2}\int_{-\pi}^{\pi} \sin(m+n)x + \sin(m-n)x\,\mathrm{d}x = 0.$$

9.4.6

(9)

$$\int_0^{\pi/2} \frac{\sin^2 nx}{\sin^2 x}\,\mathrm{d}x = \int_0^{\pi/2} \sum_{k=1}^{n} \frac{\sin(2k-1)\pi}{\sin x}\,\mathrm{d}x = \int_0^{\pi/2} \sum_{k=1}^{n}\sum_{j=1}^{k} \sin jx\,\mathrm{d}x = \frac{n\pi}{2}.$$

9.4.7

Suppose $f \in C([0,1])$, prove that

$$\int_0^{\pi} xf(\sin x)\,\mathrm{d}x = \frac{\pi}{2}\int_0^{\pi} f(\sin x)\,\mathrm{d}x,$$

and calculate $\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x}\,\mathrm{d}x$.
Proof: Note that $(\pi - x)f(\sin(\pi - x)) + xf(\sin x) = \pi f(\sin x)$, hence the identity holds.

$$\int_0^{\pi} x \cdot \frac{\sin x}{2-\sin^2 x}\,\mathrm{d}x = \frac{\pi}{2}\int_0^{\pi} \frac{\sin x}{1+\cos^2 x}\,\mathrm{d}x = \frac{\pi^2}{4}.$$

9.4.8

Suppose f is monotonically decreasing on $(0,\infty)$, denote

$$F(x) = \frac{1}{x}\int_0^x f\,\mathrm{d}t, \forall x > 0$$

Prove that F is also decreasing on $(0,\infty)$.
Proof: For any $x < x+h$,

$$x(x+h)(F(x) - F(x+h)) = h\int_0^x f\,\mathrm{d}x - x\int_x^{x+h} f\,\mathrm{d}x > hxf(x) - hxf(x) = 0.$$

Hence $F(x) > F(x+h)$.

9.4.9

Suppose f is of period T and $f \in \mathcal{R}(0,T)$, prove that

$$\lim_{\lambda\to\infty} \frac{1}{\lambda}\int_0^{\lambda} f\,\mathrm{d}x = \frac{1}{T}\int_0^T f\,\mathrm{d}x.$$

Proof: Suppose $M = \int_0^T |f|\,\mathrm{d}x, I = \int_0^T f\,\mathrm{d}x$, and let $n_\lambda = \lfloor \lambda/T \rfloor$, then

$$\left|\frac{1}{\lambda}\int_0^{\lambda} f\,\mathrm{d}x - \frac{I}{T}\right| \leqslant \left|\frac{In_\lambda}{\lambda} - \frac{I}{T}\right| + \frac{M}{\lambda} \leqslant \frac{I+M}{\lambda} \to 0.$$

Hence

$$\lim_{\lambda\to\infty} \frac{1}{\lambda}\int_0^{\lambda} f\,\mathrm{d}x = \frac{1}{T}\int_0^T f\,\mathrm{d}x.$$

9.4.10

Suppose $f \in C^2([a,b])$, prove that

$$\int_a^b f \,\mathrm{d}x = \frac{1}{2}(f(a)+f(b))(b-a) + \frac{1}{2}\int_a^b (x-a)(x-b)f''(x)\,\mathrm{d}x.$$

Proof: By integration by parts,

$$\begin{aligned}\int_a^b (x-a)(x-b)f''\,\mathrm{d}x &= -\int_a^b f'\,\mathrm{d}(x-a)(x-b) = -\int_a^b (2x-a-b)\,\mathrm{d}f \\ &= 2\int_a^b f\,\mathrm{d}x - (f(a)+f(b))(a-b).\end{aligned}$$

9.4.11

Suppose $f \in C([0,1])$, prove that

$$\lim_{n\to\infty} n\int_0^1 x^n f\,\mathrm{d}x = f(1).$$

Proof: Note that

$$\begin{aligned}\left|f(1) - (n+1)\int_0^1 x^n f(x)\,\mathrm{d}x\right| &\leqslant (n+1)\int_0^{1-\delta} x^n|f(x)-f(1)|\,\mathrm{d}x + (n+1)\int_{1-\delta}^1 x^n|f(x)-f(1)|\,\mathrm{d}x \\ &\leqslant 2M(1-\delta)^{n+1} + (1-(1-\delta)^{n+1})\varepsilon(\delta).\end{aligned}$$

Let $n\to\infty$ we know the limsup of LHS is less then $\varepsilon(\delta) = \sup_{x\in[1-\delta,1]}|f(x)-f(1)|$ for all $\delta > 0$. Since $\varepsilon(\delta)\to 0$ as $\delta \to 0$,

$$\lim_{n\to\infty} n\int_0^1 x^n f\,\mathrm{d}x = f(1).$$

9.4.12

Suppose $f \in \mathcal{R}([0,1])$, prove that

$$\lim_{n\to\infty}\int_0^1 \sqrt[n]{x} f(x)\,\mathrm{d}x = \int_0^1 f\,\mathrm{d}x.$$

Proof: Suppose $|f|$ is bounded by M, then

$$\left|\int_0^1 \sqrt[n]{x} f(x)\,\mathrm{d}x - \int_0^1 f\,\mathrm{d}x\right| \leqslant M\int_0^1 1-\sqrt[n]{x}\,\mathrm{d}x = M\left(1-\frac{n}{n+1}\right) \to 0.$$

Hence

$$\lim_{n\to\infty}\int_0^1 \sqrt[n]{x} f(x)\,\mathrm{d}x = \int_0^1 f\,\mathrm{d}x.$$

9.4.13

Prove that

$$\int_0^1 \frac{\mathrm{d}x}{1+x^n} = 1 - \frac{\log 2}{n} + o\left(\frac{1}{n}\right).$$

Proof: Note that

$$1 - \int_0^1 \frac{\mathrm{d}x}{1+x^n} = \int_0^1 \frac{x^n}{1+x^n}\,\mathrm{d}x = \frac{1}{n}\int_0^1 x\,\mathrm{d}\log(1+x^n) = \frac{\log 2}{n} - \frac{1}{n}\int_0^1 \log(1+x^n)\,\mathrm{d}x$$

And

$$\int_0^1 \log(1+x^n)\,\mathrm{d}x \leqslant \int_0^1 x^n\,\mathrm{d}x = \frac{1}{n+1}.$$

Hence

$$\int_0^1 \frac{\mathrm{d}x}{1+x^n} = 1 - \frac{\log 2}{n} + o\left(\frac{1}{n}\right).$$

9.4.14

Suppose $f \in C([-1,1])$, prove that

$$\lim_{h\to 0^+} \int_{-1}^1 \frac{hf(x)}{h^2+x^2}\,\mathrm{d}x = \pi f(0).$$

Proof: Let $K_h(x) = h/(h^2+x^2)$ then

$$\int_{-1}^1 K_h(x)\,\mathrm{d}x = 2\arctan\frac{1}{h} \to \pi,$$

and

$$\int_{|x|>\delta} K_h(x)\,\mathrm{d}x < \pi - 2\arctan\frac{\delta}{h}.$$

Hence K_h is sort of a good kernel. Note that

$$\begin{aligned}\left|\int_{-1}^1 \frac{hf(x)}{h^2+x^2}\,\mathrm{d}x - \pi f(0)\right| &\leqslant \left(\pi - 2\arctan\frac{1}{h}\right)|f(0)| + \int_{-\delta}^{\delta} K_h(x)|f(x)-f(0)|\,\mathrm{d}x \\ &\quad + \int_{|x|>\delta} K_h(x)|f(x)-f(0)|\,\mathrm{d}x \\ &\leqslant \left(\pi - 2\arctan\frac{1}{h}\right)|f(0)| + \pi\varepsilon(\delta) + 2M\left(\pi - 2\arctan\frac{\delta}{h}\right).\end{aligned}$$

Where $\varepsilon(\delta) = \sup_{x\in[-\delta,\delta]}|f(x)-f(0)|$ and $M = \sup_{x\in[-1,1]}|f(x)|$. Let $\delta = \sqrt{h}$ then the right hand side tends to zero as $h \to 0^+$. Therefore

$$\lim_{h\to 0^+}\int_{-1}^{1}\frac{hf(x)}{h^2+x^2}\,\mathrm{d}x=\pi f(0).$$

9.5

9.5.2

Suppose $f\in\mathcal{R}([a,b])$, and let $F(x)=\int_a^x f(t)\,\mathrm{d}t$, prove that

$$\int_a^b F(x)\,\mathrm{d}x=\int_a^b (b-x)f(x)\,\mathrm{d}x.$$

Proof: We show that $\int_a^x F(t)\,\mathrm{d}t=\int_a^x (x-t)f(t)\,\mathrm{d}t$ for all $x\in[a,b]$: the case $x=a$ is trivial. Taking derivatives on both sides we obtain $F(x)=\int_a^x f(t)\,\mathrm{d}t$ which is clearly true. Hence the identity holds.

9.5.4

Prove that
(1) $\lim_{n\to\infty}\int_0^{\pi/2}\sin^n x\,\mathrm{d}x=0$:

$$\left|\int_0^{\pi/2}\sin^n x\,\mathrm{d}x\right|\leqslant\frac{\pi}{2}\sin^n\left(\frac{\pi}{2}-\delta\right)+\delta\to\delta$$

for all $\delta>0$, hence $\int_0^{\pi/2}\sin^n x\,\mathrm{d}x\to 0$.
(2) $\lim_{n\to\infty}\int_n^{n+p}\frac{\cos x}{x}\,\mathrm{d}x=0$:

$$\left|\int_n^{n+p}\frac{\cos x}{x}\,\mathrm{d}x\right|\leqslant\int_n^{n+p}\frac{1}{x}\,\mathrm{d}x=\log\left(1+\frac{p}{n}\right)\to 0.$$

9.5.6

Suppose $f(x)=\int_x^{x+1}\sin t^2\,\mathrm{d}t$, prove that $|f(x)|<\frac{1}{x}$ when $x>0$.
Proof: Note that

$$f(x)=\int_x^{x+1}\frac{-1}{2t}\,\mathrm{d}\cos t^2=\frac{\cos x^2}{2x}-\frac{\cos(x+1)^2}{2(x+1)}-\int_x^{x+1}\frac{\cos t^2}{2t^2}\,\mathrm{d}t.$$

Hence

$$|f(x)|<\frac{1}{2x}+\frac{1}{2(x+1)}+\frac{1}{2x}-\frac{1}{2(x+1)}=\frac{1}{x}.$$

9.5.8

Suppose $f\in C([a,b])$, prove that

$$\lim_{n\to\infty}\left(\int_a^b|f|^n\,\mathrm{d}x\right)^{1/n}=\max_{x\in[a,b]}|f(x)|.$$

Proof: Clearly $\left(\int_a^b |f|^n \,\mathrm{d}x\right)^{1/n} \leqslant \max_{x\in[a,b]}|f(x)|$. Suppose $f(x_0) = M = \sup_{x\in[a,b]}|f(x)|$ and $x_0 \neq b$, for any $\varepsilon > 0$ take $\delta > 0$ such that $f(y) > M - \varepsilon \forall y \in [x_0, x_0 + \delta]$, then

$$\left(\int_a^b |f|^n \,\mathrm{d}x\right)^{1/n} \geqslant \delta^{1/n}(M - \varepsilon) \to M - \varepsilon$$

for all $\varepsilon > 0$, hence $\lim_{n\to\infty}\left(\int_a^b |f|^n \,\mathrm{d}x\right)^{1/n} = M$.

9.5.9

Suppose $f, g \in \mathcal{R}([a,b])$ and g has period T, prove that

$$\lim_{\lambda\to\infty}\int_a^b f(x)g(\lambda x)\,\mathrm{d}x = \frac{1}{T}\int_a^b f\,\mathrm{d}x\int_0^T g\,\mathrm{d}x.$$

Proof: The case $f = \chi_{[c,d]}$ is proved in Exercise9.4.9, so the identity holds for step functions.
For any $\varepsilon > 0$, there is a step function ϕ such that $\int_a^b |f - \phi|\,\mathrm{d}x < \varepsilon$. Let $I = \int_0^T g\,\mathrm{d}x, M = \sup|g|$, then

$$\left|\int_a^b f(x)g(\lambda x)\,\mathrm{d}x - \frac{I}{T}\int_a^b f\,\mathrm{d}x\right| \leqslant \left(M + \frac{I}{T}\right)\int_a^b |f - \phi|\,\mathrm{d}x + \delta(\lambda) < \varepsilon\left(M + \frac{I}{T}\right) + \delta(\lambda).$$

where $\delta(\lambda) = \left|\int_a^b \phi(x)g(\lambda x)\,\mathrm{d}x - \frac{I}{T}\int_a^b \phi\,\mathrm{d}x\right|$. Let $\lambda \to \infty$ then the lim sup is less then $\varepsilon(M + I/T)$ for any $\varepsilon > 0$, hence it is zero and the identity holds for arbitrary f.

9.6

9.6.6

Suppose $f \in C^1([a,b])$ and $|f'| \leqslant M$, prove that

$$\left|\int_a^b f\,\mathrm{d}x - \frac{f(a) + f(b)}{2}(b - a)\right| \leqslant \frac{1}{4}M(b - a)^2.$$

Proof: Note that

$$\int_a^b f\,\mathrm{d}x - \frac{f(a) + f(b)}{2}(b - a) = -\frac{1}{2}\int_a^b (2x - a - b)f'\,\mathrm{d}x$$

and

$$\int_a^b |2x - a - b|\,\mathrm{d}x = \frac{(b - a)^2}{2}.$$

9.6.7

Suppose $f \in C^2([a,b])$, prove that there exists $\xi \in (a,b)$ such that

$$\int_a^b f\,\mathrm{d}x = (b-a)f\left(\frac{a+b}{2}\right) + \frac{1}{24}(b-a)^3 f''(\xi).$$

Proof: Let $c = \frac{a+b}{2}$ and $h = \frac{b-a}{2}$, consider

$$F(t) = \int_{c-t}^{c+t} f\,\mathrm{d}t - 2tf(c) + Kt^3$$

where K is a constant such that $F(h) = 0$. Clearly F is an odd function, and $F'(0) = 0$ so $0 = F(h) = h^3 F^{(3)}(\eta)$ for some $\eta \in (0, h)$. Hence $\xi = \eta + c$ satisfy the requirements.

9.7

9.7.1

Calculate arc length:
(1) $y = x^{3/2}, 0 \leqslant x \leqslant 4$.

$$\text{length} = \int_0^4 \sqrt{1 + \frac{9}{4}x}\,\mathrm{d}x = \frac{8}{27}(10\sqrt{10} - 1).$$

9.7.2

Calculate area:
(3) $x^2 + 9y^2 = 1$:

$$\text{Area} = \frac{1}{3}\pi.$$

9.7.3

Calculate volume:
(1) $y = \tan x, x \in [0, \pi/4]$.

$$\text{Volume} = \int_0^{\pi/4} \pi(\tan x)^2\,\mathrm{d}x = -\frac{\pi^2}{4} + \pi\int_0^{\pi/4} \mathrm{d}\tan x = \pi - \frac{\pi^2}{4}.$$

9.7.4

Calculate volume:
(1) $x + y + z^2 = 1, x = 0, y = 0$.

$$\text{Volume} = \int_{x+y\leqslant 1} 2\sqrt{1-x-y}\,\mathrm{d}x\mathrm{d}y = 2\int_0^1 (1-t)\sqrt{t}\,\mathrm{d}t = \frac{8}{15}.$$

9.7.5

Calculate volume:
(4) $y = \sin x, x \in [0, \pi]$.

$$\text{Volume} = \int_0^\pi \pi(\sin x)^2 \,\mathrm{d}x = \frac{\pi}{2}\int_0^\pi 1 - \cos 2x \,\mathrm{d}x = \frac{\pi^2}{2}.$$

9.7.10

Prove that π is irrational.
Proof: Otherwise suppose $\pi = a/b$. Let $f(x) = \frac{1}{n!}x^n(a-bx)^n$. By Lebiniz's formula, $f^{(k)}(x) = \frac{1}{n!}\sum_{j=0}^k \binom{k}{j}(x^n)^{(j)}((a-bx)^n)^{(k-j)}$ vanishes at $x = 0, a/b$ for $0 \leqslant k \leqslant n$, and for $k > n$, each term has coefficient $\binom{k}{j}j!/(n-k+j)$ which is an integer.
Using integration by parts,

$$\int_0^\pi f(x)\sin x \,\mathrm{d}x = \sum_{k=0}^n (-1)^k(f^{(2k)}(\pi) - f^{(2k)}(0)) \in \mathbb{Z}.$$

However, note that

$$0 < \int_0^\pi f(x)\sin x \,\mathrm{d}x < \sup_{x\in[0,\pi]} f(x) \leqslant \frac{(a\pi)^{2n}}{n!} \to 0.$$

Hence for large n, $\int_0^\pi f(x)\sin x \,\mathrm{d}x \notin \mathbb{Z}$, a contradiction.